OLLAMA COMMANDS AND THEIR FUNCTIONS

Introduction
Ollama is a platform that allows users to run Large Language Models (LLMs) locally on their computers. It provides simple commands to download, manage, and interact with AI models such as Llama 3, Mistral, Gemma, and DeepSeek.

Objectives
Understand the basic Ollama commands.
Learn how to download and run AI models.
Manage installed and running models.
Configure model parameters.

Commands and Descriptions
1. ollama --version
Purpose: Displays the installed Ollama version.
Command:ollama --version

2. ollama help
Purpose: Shows all available Ollama commands.
Command:ollama help

3. ollama serve
Purpose: Starts the Ollama server.
Command:ollama serve

4. ollama pull
Purpose: Downloads a model from the Ollama library.
Command:ollama pull llama3

5. ollama list
Purpose: Displays all installed models.
Command:ollama list

6. ollama run
Purpose: Runs a model and starts a chat session.
Command:ollama run llama3

7. ollama ps
Purpose: Shows currently running models.
Command:ollama ps

8. ollama show
Purpose: Displays information about a model.
Command:ollama show llama3

9. ollama stop
Purpose: Stops a running model.
Command:ollama stop llama3

10. ollama rm

Purpose: Removes a model from the system.

Command:ollama rm llama3

```
Command Prompt

Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\91831>ollama ls
NAME           ID              SIZE      MODIFIED
llama3.2:3b    a80c4f17acd5    2.0 GB    2 hours ago

C:\Users\91831>ollama run llama3.2:3b
>>> hi
How can I assist you today?

>>> /bye

C:\Users\91831>ollama ps
NAME           ID              SIZE      PROCESSOR    CONTEXT    UNTIL
llama3.2:3b    a80c4f17acd5    2.6 GB    100% CPU     4096       4 minutes from now

C:\Users\91831>ollama show llama3.2:3b
  Model
    architecture        llama
    parameters          3.2B
    context length      131072
    embedding length    3072
    quantization        Q4_K_M

  Capabilities
    completion
    tools

  Parameters
    stop    "<|start_header_id|>"
    stop    "<|end_header_id|>"
    stop    "<|eot_id|>"

  License
    LLAMA 3.2 COMMUNITY LICENSE AGREEMENT
    Llama 3.2 Version Release Date: September 25, 2024
    ...
```

USING POSTMAN TO RUN API REQUESTS

Introduction
Postman is a popular API testing tool used by developers to send requests to servers and analyze responses. It provides a user-friendly interface for testing REST APIs without writing code.

Objectives
Understand the basics of API testing.
Learn how to send GET and POST requests using Postman.
Analyze API responses.
Test local AI APIs such as Ollama.

Software Required
Postman
Internet connection (for online APIs)
Ollama (optional, for local AI API testing)

Procedure
Step 1: Open Postman
Launch the Postman application on your computer.

Step 2: Create a New Request
Click New.
Select HTTP Request.
Enter a request name and save it.

Step 3: Send a GET Request
Select GET from the request type dropdown.
Enter the API URL.
Example:
https://jsonplaceholder.typicode.com/posts/1
Click Send.

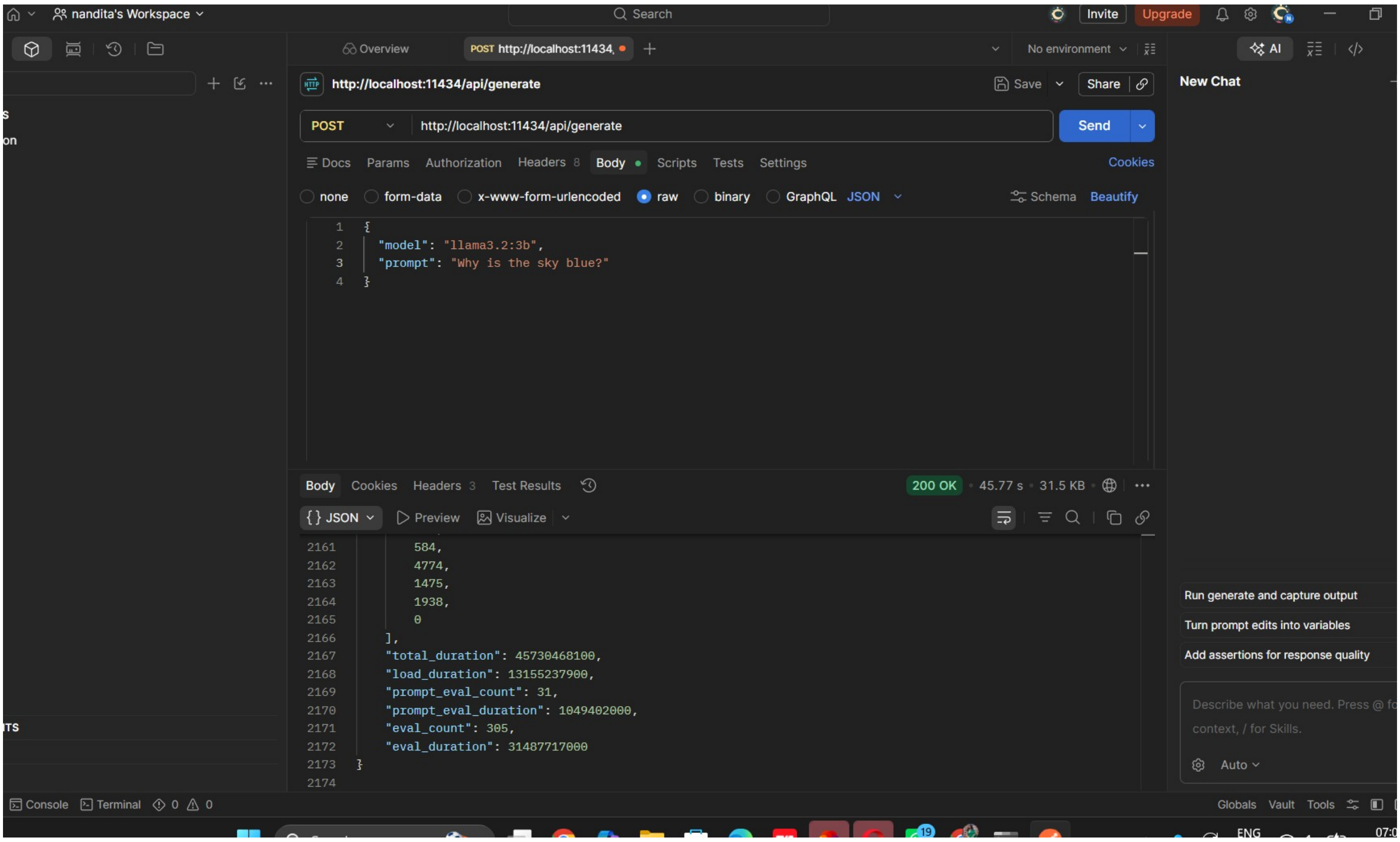
Step 4: View the Response
Postman displays:
Status Code
Response Time
Response Body (JSON)
Example Response:
JSON
```
{
  "userId": 1,
  "id": 1,
  "title": "sample title",
  "body": "sample content"
}
```

Step 5: Send a POST Request
Change the method to POST.

Enter the API URL.
Open the Body tab.
Select raw and choose JSON.
Enter JSON data.
Example:
JSON
```json
{
  "title": "Hello",
  "body": "Testing Postman"
}
```
Click Send.

Step 6: Test Ollama API
Ensure Ollama is running.
API URL:
http://localhost:11434/api/generate
Method:
POST
Body:
JSON
```json
{
  "model": "llama3",
  "prompt": "What is Artificial Intelligence?",
  "stream": false
}
```
Click Send to receive the AI-generated response.

Results
Successfully sent GET requests.
Successfully sent POST requests.
Received JSON responses from APIs.
Tested local AI models through Ollama API.
Advantages of Postman
Easy-to-use graphical interface.
Supports all HTTP methods.
Allows API automation and testing.
Displays responses in a readable format.
Useful for debugging APIs.



Conclusion

Postman is an effective tool for testing and validating APIs. It simplifies sending GET and POST requests, analyzing responses, and testing local AI APIs such as Ollama without writing code. :::
This format is suitable for a lab report, internship diary, or practical record.

API Testing Using Python Requests and Ollama

Commands

1. Basic GET Request
Python

```
import requests

response = requests.get("https://jsonplaceholder.typicode.com/posts/1")
print(response)
```

2. Check Status Code
Python

```
import requests

response = requests.get("https://jsonplaceholder.typicode.com/posts/1")
print(response.status_code)
```

3. Display JSON Data
Python

```
import requests

response = requests.get("https://jsonplaceholder.typicode.com/posts/1")
print(response.json())
```

4. Extract Title from JSON
Python

```
import requests

response = requests.get("https://jsonplaceholder.typicode.com/posts/1")
data = response.json()

print(data["title"])
```

5. Random User API
Python

```
import requests

response = requests.get("https://randomuser.me/api/")
print(response.json())
```

6. Dog API
Python

```
import requests

response = requests.get("https://dog.ceo/api/breeds/image/random")
print(response.json())
```

7. Ollama POST Request
Python

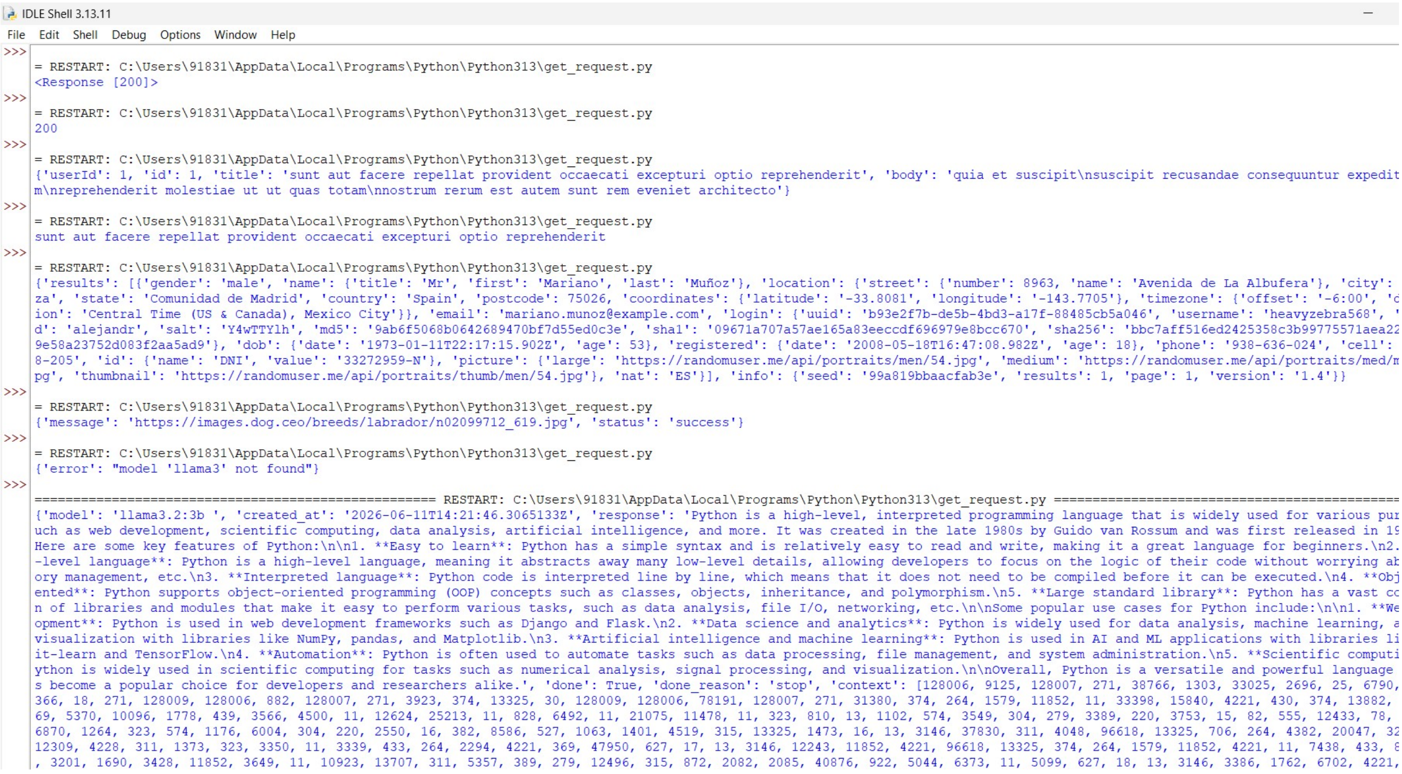
```
import requests
```

```
url = "http://localhost:11434/api/generate"

data = {
    "model": "llama3",
    "prompt": "What is Python?",
    "stream": False
}

response = requests.post(url, json=data)

print(response.json())
```



Using Ollama in Python

Step 1: Open Command Prompt
Check whether Ollama is installed:
ollama --version

Step 2: View Available Models
ollama list

Step 3: Download a Model (if not already installed)
ollama pull llama3

Step 4: Start Ollama
ollama serve

Keep this Command Prompt window running.

Step 5: Install Ollama Python Library

Open another Command Prompt:
pip install ollama

Step 6: Create a Python File
Create a file named:
ollama_test.py

Step 7: Import Ollama
Python
```
import ollama
```

Step 8: Generate a Response
Python
```
import ollama

response = ollama.generate(
   model='llama3',
   prompt='What is Python?'
)

print(response['response'])
```

Step 9: Save the File
Save as:
ollama_test.py

Step 10: Run the Program
python ollama_test.py
Output:
Python is a high-level programming language…

Step 11: Chat Example
Python
```
import ollama

response = ollama.chat(
   model='llama3',
   messages=[
      {
         'role': 'user',
         'content': 'Hello'
      }
   ]
)

print(response['message']['content'])
```

Step 12: List Models in Python
Python
```
import ollama

print(ollama.list())
```

IDLE Shell 3.13.11

File Edit Shell Debug Options Window Help

```
Python 3.13.11 (tags/v3.13.11:6278944, Dec  5 2025, 16:26:58) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:\Users\91831\AppData\Local\Programs\Python\Python313\get_request.py
Python is a high-level, interpreted programming language that is widely used for various purposes such as web development, scientific computing, data analysis, artificial intelligence,
and more. It was created in the late 1980s by Guido van Rossum and was first released in 1991.

Here are some key features of Python:

1. **Easy to learn**: Python has a simple syntax and is relatively easy to read and write, making it a great language for beginners.
2. **High-level language**: Python is a high-level language, meaning that it abstracts away many low-level details, allowing developers to focus on the logic of their program without w
orrying about memory management or other details.
3. **Interpreted language**: Python code is interpreted by an interpreter at runtime, rather than being compiled into machine code beforehand.
4. **Object-oriented**: Python supports object-oriented programming (OOP) concepts such as classes, objects, inheritance, and polymorphism.
5. **Large standard library**: Python has a vast collection of built-in modules and libraries that can be used for various tasks, including file I/O, networking, data structures, and m
ore.
6. **Cross-platform**: Python can run on multiple operating systems, including Windows, macOS, and Linux.

Some common uses of Python include:

1. **Web development**: Python is often used with frameworks like Django and Flask to build web applications.
2. **Data analysis**: Python has libraries like NumPy, pandas, and matplotlib that make it easy to analyze and visualize data.
3. **Artificial intelligence and machine learning**: Python is widely used in AI and ML applications due to its simplicity and extensive libraries like scikit-learn and TensorFlow.
4. **Automation**: Python can be used to automate tasks, such as data scraping, file management, and more.
5. **Scientific computing**: Python has extensive support for scientific computing, including libraries like NumPy and SciPy.

Some of the key benefits of using Python include:

1. **Fast development**: Python's syntax is simple and easy to read, making it ideal for rapid prototyping and development.
2. **Large community**: Python has a massive and active community of developers who contribute to its ecosystem and provide support.
3. **Extensive libraries**: Python's standard library and third-party libraries make it easy to find solutions to common problems.

Overall, Python is a versatile and powerful language that is widely used in various industries and applications.
>>>
```